

Python Intermediate Day 3 Checkpoint

- Operator Overloading
- Designing with Classes
- Advanced Class Topics
- Decorators & Class Gotchas
- Software Engineering term
- Best practice of development with LLM

1. Email *

2. Name

3. If you want to support the + operator for a custom class, you must implement:

1 point

Mark only one oval.

- ☐ A. __plus__
- ☐ B. __add__
- ☐ C. __sum__
- ☐ D. __concat__

4. Which method enables indexing like `obj[0]`?

1 point

Mark only one oval.

- ☐ A. `__index__`
- ☐ B. `__getitem__`
- ☐ C. `__access__`
- ☐ D. `__slice__`

5. To override the `<` operator, you implement:

1 point

Mark only one oval.

- ☐ A. `__less__`
- ☐ B. `__lt__`
- ☐ C. `__cmp__`
- ☐ D. `__compare__`

6. What is the main advantage of composition over inheritance?

1 point

Mark only one oval.

- ☐ A. Faster execution
- ☐ B. Less memory usage
- ☐ C. Better flexibility and loose coupling
- ☐ D. Automatic polymorphism

7. “Is-a” relationship represents:

1 point

Mark only one oval.

- ☐ A. Composition
- ☐ B. Delegation
- ☐ C. Inheritance
- ☐ D. Factory pattern

8. A mix-in class is primarily used to:

1 point

Mark only one oval.

- ☐ A. Replace inheritance
- ☐ B. Provide reusable features across classes
- ☐ C. Speed up execution
- ☐ D. Prevent subclassing

9. MRO stands for:

1 point

Mark only one oval.

- ☐ A. Multiple Resolution Operator
- ☐ B. Method Resolution Order
- ☐ C. Module Runtime Override
- ☐ D. Memory Reference Order

10. Which decorator turns a method into a class-level method?

1 point

Mark only one oval.

- ☐ A. @staticmethod
- ☐ B. @property
- ☐ C. @classmethod
- ☐ D. @decorator

11. Which feature allows attribute validation logic?

1 point

Mark only one oval.

- ☐ A. Mixins
- ☐ B. Descriptors / Properties
- ☐ C. MRO
- ☐ D. Metaclasses

12. What is the purpose of a factory pattern?

1 point

Mark only one oval.

- ☐ A. Enforce inheritance
- ☐ B. Centralize object creation logic
- ☐ C. Replace constructors
- ☐ D. Prevent subclassing

13. In UML, inheritance is represented by:

1 point

Mark only one oval.

- ☐ A. Dashed arrow
- ☐ B. Solid line with hollow triangle
- ☐ C. Double arrow
- ☐ D. Diamond shape

14. In object design, “cross-cutting concerns” include:

1 point

Mark only one oval.

- ☐ A. Data structures
- ☐ B. Logging and authentication
- ☐ C. Inheritance trees
- ☐ D. Class variables

15. Why prefer getters and setters (or properties)?

1 point

Mark only one oval.

- ☐ A. Slower execution
- ☐ B. Direct attribute access is unsafe
- ☐ C. Encapsulation and validation control
- ☐ D. Required by Python

16. RAG (Retrieval-Augmented Generation) combines:

1 point

Mark only one oval.

- ☐ A. Database joins and APIs
- ☐ B. Pre-trained LLM with external knowledge retrieval
- ☐ C. Reinforcement learning only
- ☐ D. Random token generation

17. LLMs primarily:

1 point

Mark only one oval.

- ☐ A. Understand meaning like humans
- ☐ B. Execute Python code
- ☐ C. Predict the next most probable token
- ☐ D. Store knowledge permanently

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